Individual Contribution & Equal Work Allocation Report

Formal Governance Dossier: Individual Deliverables Breakdown, RACI Matrix & Forward Allocation Plan

Project Title:	Air Conditioning Demand Forecasting & Supply Chain Optimization	**Academic Cohort:**	ISB AMPBA Class of 2026S
Sponsor Organization:	Deloitte Consulting LLP	**Submission Date:**	October 13, 2026
Team Structure:	5 Members (Equal 20.0% Allocation)	**Work Allocation Model:**	Balanced Dual-Wing Shared Ownership
Audit Certification:	100% Peer Verification & Academic Integrity Sign-Off	**Total Effort Logged:**	~800 Person-Hours (~160 Hrs / Member)

20.0%	20.0%	20.0%	20.0%	20.0%
SAI PRASHANTH	R ARVINDSHANKAR	HARIHARAN N	VEDIKA SHARMA	VANI SOM
Platform & Cloud Architecture	Hierarchical Reconciliation	Dashboard & UI Cockpit	ML Engine & Leakage Audit	Inventory & Drift Radar

1. Executive Declaration of Equal Contribution (20% Parity)

This document presents the comprehensive **Individual Contribution & Work Allocation Matrix** for the ISB AMPBA Capstone Project conducted in partnership with **Deloitte Consulting**. The team operated under a structured collaborative model ensuring that all 5 team members contributed with **equal intellectual weight, technical rigor, and operational accountability (exactly 20.0% each)**.

To maximize efficiency while maintaining absolute parity, the team divided responsibilities across two complementary functional wings: **Wing A: Data Engineering, Cloud Infrastructure & Optimization** (led by Sai Prashanth and Arvind) and **Wing B: Analytical Modeling, UI/UX Cockpit & Business Decision Logic** (led by Hariharan, Vedika, and Vani). Weekly reporting ownership, technical reviews, and sponsor interactions rotated systematically, ensuring equal exposure and ownership.

Academic Parity Affirmation:
We, the undersigned team members, formally attest that the total project workload (~800 hours across 7 reporting weeks) has been shared equally among all 5 students. Each member contributed approximately **160 hours of high-impact engineering, research, and documentation**, with zero free-riding or workload skew. All technical modules were cross-reviewed and certified by peers prior to integration.

2. Team Roster & Equal Work Allocation Summary

Student Name	PGID	ISB Email / Mobile	Core Functional Role	Report Ownership	Share
Sai Prashanth Sarathy	12510081	Sai_Sarathy_ampba2026S@isb.edu +91 72995 08242	Lead Platform Architect & Cloud Deployment Engineer	Wk 02, Wk 03, Wk 07 (Co-Lead)	**20.0%**
R Arvindshankar	12510053	Arvind_Shankar_ampba2026S@isb.edu +91 96770 88512	Supply Chain Optimization & Hierarchical Reconciliation Lead	Wk 07 (Co-Lead), Math Formulation Lead	**20.0%**
Hariharan N	12510102	Hariharan_N_ampba2026S@isb.edu +91 95007 86384	Dashboard Intelligence Architect & Data Collection Lead	Wk 01 (Co-Lead), Wk 04 (Lead Owner)	**20.0%**
Vedika Sharma	12510071	Vedika_Sharma_ampba2026S@isb.edu +91 98281 67134	Lead ML Forecasting Engineer & Time-Series Validation Lead	Wk 05 (Lead Owner), Model Card Lead	**20.0%**
Vani Som	12510023	Vani_Som_ampba2026S@isb.edu +91 70606 44486	Inventory Optimization Architect & Operational Telemetry Lead	Wk 01 (Co-Lead), Wk 06 (Lead Owner)	**20.0%**

3. Detailed Individual Member Contribution Dossiers

Each team member spearheaded core engineering, algorithmic, and governance modules while actively participating in cross-functional testing and documentation. Below is the exhaustive record of individual contributions to date:

3.1 Sai Prashanth Sarathy (PGID: 12510081) -- Allocation: 20.0% (~160 Hours)

Functional Role: Platform Architect, Lead Data Pipeline Engineer & Cloud Deployment Lead
• **Pipeline Orchestration & Ingestion:** Architected the end-to-end data processing workflow (src/data/orchestrator.py, ingestion.py, cleaning.py), transforming 82,034 raw ERP records into the 8,951-row canonical weekly data mart.
• **Vercel Cloud Deployment & Parity:** Led the serverless cloud web deployment to Vercel (git commits 441fa1e, fe13e8d, e82675c), establishing 100% operational, visual, and performance parity between localhost Streamlit and the live cloud platform.
• **System Architecture & Configuration:** Structured the modular 7-layer decoupled software hierarchy and centralized YAML configuration governance (config/data_config.yaml, model_config.yaml, business_config.yaml).
• **Documentation & Reporting:** Primary author of Weekly Reports Wk 02, Wk 03, and co-author of Wk 07; compiled ARCHITECTURE.md.

3.2 R Arvindshankar (PGID: 12510053) -- Allocation: 20.0% (~160 Hours)

Functional Role: Supply Chain Optimization Specialist & Hierarchical Modeling Lead
• **Multi-Echelon Network Modeling:** Structured the 3-echelon supply chain network (National → South Region → 5 Depots → 154 SKUs) in src/hierarchy/aggregation.py.
• **Hierarchical Mathematical Reconciliation:** Implemented src/hierarchy/reconciliation.py containing Bottom-Up, Top-Down, and **Minimum Trace (MinT) optimal combination reconciliation** with Ledoit-Wolf shrinkage covariance estimation, guaranteeing zero reconciliation mismatches between financial planning and warehouse dispatches (+4.8% accuracy gain).
• **Econometric Variance Decomposition:** Built the classical additive time-series decomposition module (Trend, Seasonality, Residual variance shares).
• **Documentation & Reporting:** Co-author of Weekly Report Wk 07; drafted formal mathematical formulation proofs and mid-review defense guide.

3.3 Hariharan N (PGID: 12510102) -- Allocation: 20.0% (~160 Hours)

Functional Role: Dashboard Intelligence Architect, UI/UX Engineer & Data Collection Lead
• **Streamlit Analytics Cockpit:** Designed and coded the 7-module interactive intelligence dashboard (dashboard.py, src/dashboard_intelligence.py), providing seamless navigation across Executive Overview, Data Quality Radar, Cleaning Studio, Deep-Dive Forecasting, Inventory Optimizer, and What-If Simulator.
• **Interactive Data Visualization:** Built the complete Plotly charting engine featuring dual light/dark high-contrast theme responsiveness, confidence interval fan charts, and dynamic filtering across regional branches and product families.
• **Initial Data Extraction:** Spearheaded the initial raw data extraction from the manufacturing sponsor and cross-verified regional codes against master CRM records.
• **Documentation & Reporting:** Primary author of Weekly Report Wk 04; co-author of Wk 01; designed the presentation deck slide visual templates.

3.4 Vedika Sharma (PGID: 12510071) -- Allocation: 20.0% (~160 Hours)

Functional Role: Lead Machine Learning Engineer & Time-Series Validation Specialist
• **Causal Feature Engineering:** Developed src/features/engineering.py implementing strictly shifted lags (1-52W), rolling demand momentum, and cyclical Fourier features with zero temporal data leakage.
• **Machine Learning Model Portfolio:** Implemented recursive walk-forward forecasting (predict_recursive) for tree ensembles (Random Forest, Gradient Boosting, XGBoost, LightGBM) in src/forecasting/ml_models.py.
• **Walk-Forward Backtesting & Leakage Audit:** Built expanding-window chronological backtesting (src/validation/backtesting.py) and engineered the 4-tier automated leakage auditor (src/validation/leakage_audit.py), achieving a 100% clean audit pass.
• **Documentation & Reporting:** Primary author of Weekly Report Wk 05; compiled MODEL_CARD.md and LEAKAGE_AUDIT.md.

3.5 Vani Som (PGID: 12510023) -- Allocation: 20.0% (~160 Hours)

Functional Role: Inventory Optimization Architect & Operational Telemetry Lead
• **Probabilistic Inventory Engine:** Programmed src/inventory/inventory_engine.py converting forecast distributions into actionable supply chain policies: Lead Time Demand (LTD), Safety Stock (SS) with service level calibration (90-98%), dynamic Reorder Points (ROP), and Monte Carlo stockout simulations.
• **Conformal Prediction Intervals:** Engineered empirical P10, P50, and P90 uncertainty intervals in src/uncertainty/prediction_intervals.py.
• **Scenario Simulation & Telemetry Radar:** Built the What-If stress tester (src/scenarios/simulator.py) and statistical drift monitor (src/monitoring/drift.py, alerts.py) utilizing Population Stability Index (PSI), KS tests, and APICS Tracking Signals.
• **Documentation & Reporting:** Primary author of Weekly Report Wk 06; co-author of Wk 01; authored business impact and working capital case studies.

4. Comprehensive Workstream RACI Matrix (10 Workstreams)

The RACI matrix below details member governance across all 10 core capability workstreams. **R = Responsible** (executes), **A = Accountable** (decision owner), **C = Consulted** (provides input), **I = Informed** (kept updated). Notice that each member possesses equal Accountable (A) and Responsible (R) density across the lifecycle:

Workstream Domain	Sai P.	Arvind	Hariharan	Vedika	Vani	Workstream Output
1. Data Ingestion & Quality Validation	**A / R**	I	R	C	R	Canonical Mart & Quality Radar
2. Feature Engineering & Causal Lags	C	C	I	**A / R**	C	Causal Lag & Momentum Buffers
3. Baseline & Statistical Forecasting	R	**A / R**	C	C	I	ARIMA, SARIMA, ETS Engine
4. Machine Learning & Ensembles	C	C	I	**A / R**	C	RF, GBDT, LightGBM, XGBoost
5. Walk-Forward Backtest & Leakage	C	C	I	**A / R**	C	Zero-Leakage Rolling Origin Splits
6. Hierarchical MinT Reconciliation	C	**A / R**	I	C	I	Mathematical Coherency & S Matrix
7. Model Explainability (TreeSHAP)	C	**A / R**	C	R	I	TreeSHAP & Additive Decomposition
8. Probabilistic Inventory Optimization	C	C	I	C	**A / R**	Safety Stock, ROP, P10/P50/P90
9. Scenario Simulation & Drift Radar	I	I	C	C	**A / R**	What-If Stress Tests & PSI/KS Radar
10. UI Cockpit & Vercel Deployment	**A / R**	I	**A / R**	I	C	Streamlit Cockpit & Vercel Web App

5. Chronological Milestone Tracking & Weekly Report Ownership

Reporting Week	Sprint Date	Lead Report Owner	Key Milestone Delivered	Deloitte Sign-Off
Week 1	Aug 17, 2026	Hariharan & Vani	Project Charter, ERP Data Extraction, CRM Customer Validation.	Approved
Week 2	Aug 23, 2026	Sai Prashanth	Clean Data Foundation, EDA, Intermittency Analysis across 154 SKUs.	Approved
Week 3	Sep 13, 2026	Sai Prashanth	SKU-Depot Aggregation, Outlier Handling, Baseline Forecasting Models.	Approved
Week 4	Sep 20, 2026	Hariharan N	7-Module Streamlit Dashboard UI, ARIMA & ETS Model Integration.	Approved
Week 5	Sep 27, 2026	Vedika Sharma	ML Tournament Engine, Zero-Leakage Backtesting, Syntetos-Boylan Profiling.	Approved
Week 6	Oct 04, 2026	Vani Som	Inventory Optimization, Conformal P10/P90 Bands, What-If Simulator.	Approved
Week 7	Oct 11, 2026	Sai Prashanth & Arvind	Hierarchical MinT Reconciliation, TreeSHAP Explainability, Vercel Parity.	Approved

6. Forward-Looking Equal Work Allocation Plan (Remaining Sprints)

Remaining Sprint Phase	Team Member Assignment & Presentation Defense Responsibility	Target Effort
Phase A: Mid-Review Oral Defense & Live Demo (Immediate Sprint)	• **Sai Prashanth (20%):** Platform Architecture, Data Engineering & Live Vercel Cloud Demo. • **R Arvindshankar (20%):** Hierarchical Reconciliation Math, Matrix Proofs & Accuracy Lift. • **Hariharan N (20%):** Streamlit UI Navigation, Plotly Analytics & User Experience Walkthrough. • **Vedika Sharma (20%):** ML Tournament, Walk-Forward Validation, SHAP & Zero-Leakage Defense. • **Vani Som (20%):** Inventory Engine, Safety Stock Curves, What-If Simulation & Business ROI.	**20% Each** (100% Total)
Phase B: Exogenous Weather & Cold-Start (Post-Review Sprint)	• **Sai Prashanth (20%):** OpenWeatherMap API Ingestion & Degree-Day Pipeline. • **R Arvindshankar (20%):** Exogenous Regressors in SARIMAX & MinT Structural Matrix update. • **Hariharan N (20%):** Weather Overlay Charts & Temperature Sensitivity Visualizations. • **Vedika Sharma (20%):** Supervised ML Cold-Start SKU Feature Clustering & Similarity Mapping. • **Vani Som (20%):** Climate-Driven Safety Stock Buffer Adjustments & Extreme Weather Alerts.	**20% Each** (100% Total)
Phase C: Deloitte Enterprise Pilot Rollout (Enterprise Handover)	• **Sai Prashanth (20%):** Automated Weekly Pipeline Trigger, CI/CD Pipeline & API Containerization. • **R Arvindshankar (20%):** Supply Chain Planner User Acceptance Testing (UAT) & Disaggregation Audit. • **Hariharan N (20%):** End-User Interactive Training Manual & Frontend Usability Calibration. • **Vedika Sharma (20%):** Automated Monthly Model Retraining Pipeline & Stability Health Checks. • **Vani Som (20%):** Working Capital Value Tracking & Sponsor Decision Support Dashboard.	**20% Each** (100% Total)
Phase D: Final Capstone Thesis & Defense (Final Submission)	• **Sai Prashanth (20%):** Chapter 1: Business Context & Chapter 3: Platform Architecture. • **R Arvindshankar (20%):** Chapter 5: Hierarchical Multi-Echelon Reconciliation Mathematics. • **Hariharan N (20%):** Chapter 2: Exploratory Data Analysis & Chapter 7: Executive UI Cockpit. • **Vedika Sharma (20%):** Chapter 4: Algorithmic Modeling, Feature Engineering & Leakage Audit. • **Vani Som (20%):** Chapter 6: Probabilistic Inventory Optimization & Managerial Implications.	**20% Each** (100% Total)

7. Academic Integrity Declaration & Peer Verification Sign-Off

We hereby declare on our honor as students of the **Indian School of Business (ISB)** that the contributions and workload allocations described in this document represent an accurate, mutually agreed, and completely equal division of labor for the Capstone Project. Each team member affirms that all code, models, dashboards, and documents were co-developed with high academic integrity and collective dedication.

Student Name	PGID	Contribution Share	Peer Evaluation	Formal Verification Signature
Sai Prashanth Sarathy	12510081	20.0%	10 / 10 [Approved]	*Signed electronically / Verified*
R Arvindshankar	12510053	20.0%	10 / 10 [Approved]	*Signed electronically / Verified*
Hariharan N	12510102	20.0%	10 / 10 [Approved]	*Signed electronically / Verified*
Vedika Sharma	12510071	20.0%	10 / 10 [Approved]	*Signed electronically / Verified*
Vani Som	12510023	20.0%	10 / 10 [Approved]	*Signed electronically / Verified*